



RLHF and VLMs

Course AIAA 4220

Week 11- Lecture 22

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Limitation of Prompting

- **Language models are not aligned with user intent**

PROMPT	<i>Explain the moon landing to a 6 year old in a few sentences.</i>
COMPLETION	GPT-3 Explain the theory of gravity to a 6 year old. Explain the theory of relativity to a 6 year old in a few sentences. Explain the big bang theory to a 6 year old. Explain evolution to a 6 year old.

The big pretrained model just fails at some cases. We don't want to go back and do the pretraining again. Then how?

Limitation of Prompting

- Let's say we were training a language model on some task (e.g. summarization)
- For each LLM sample s , imagine we had a way to obtain a human reward of that summary: $R(s) \in \mathbb{R}$, higher is better
- Now we want to **maximize the expected reward of samples from our LLM:**

$$\mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})]$$

SAN FRANCISCO,
California (CNN) --
A magnitude 4.2
earthquake shook the
San Francisco
...
overturn unstable
objects.

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$s_1 \\ R(s_1) = 8.0$$

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

$$s_2 \\ R(s_2) = 1.2$$

Optimizing for human preferences

- How do we actually change our LLM parameters θ to maximize this?

$$\mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})]$$

- Let's try doing gradient ascent!

$$\theta_{t+1} := \theta_t + \alpha \nabla_{\theta_t} \mathbb{E}_{\hat{s} \sim p_{\theta_t}(s)}[R(\hat{s})]$$

How do we estimate this expectation?

What if our reward function is nondifferentiable?

- **Policy gradient methods in RL** (e.g., REINFORCE; [Williams, 1992]) give us tools for estimating and optimizing this objective

Policy Gradient/REINFORCE

- We can approximate this objective with Monte Carlo samples
 - Remember our pong example: we simply say all actions for game won are positive and vice versa

Maximizing this: $\nabla_{\theta} \mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})] = \mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s}) \nabla_{\theta} \log p_{\theta}(\hat{s})] \approx \frac{1}{m} \sum_{i=1}^m R(s_i) \nabla_{\theta} \log p_{\theta}(s_i)$

- Giving us the update rule:

$$\theta_{t+1} := \theta_t + \alpha \frac{1}{m} \sum_{i=1}^m R(s_i) \nabla_{\theta_t} \log p_{\theta_t}(s_i)$$

s a lot

If R is ---

Take gradient steps
to maximize $p_{\theta}(s_i)$

Take steps to
minimize $p_{\theta}(s_i)$

This is why it's called "reinforcement learning": we reinforce good actions, increasing the chance they happen again.

Modelling human preferences

- Awesome: now for any arbitrary, non-differentiable reward function $R(s)$, we can train our language model to maximize expected reward
- Not so fast! (Why not?)
- Problem 1: human-in-the-loop is expensive!
 - Solution: instead of directly asking humans for preferences, model their preferences as a separate (NLP) problem!

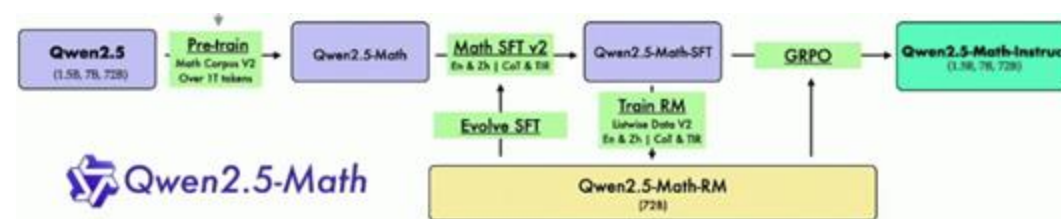
An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$s_1$$
$$R(s_1) = 8.0$$


The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

$$s_2$$
$$R(s_2) = 1.2$$


Train an LM $RM_\phi(s)$ to predict human preferences from an annotated dataset, then optimize for RM_ϕ instead



Modelling human preferences

- Problem 2: human judgments are noisy and miscalibrated!

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

s_3

$R(s_3) = 4.1? \quad 6.6? \quad 3.2?$

It is difficult to get consistent absolute score evaluation from humans.

Solution: instead of asking for direct ratings, ask for pairwise comparisons, which can be more reliable

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

s_1

>

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

s_3

>

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

s_2

RLHF (Reinforcement Learning from Human Feedback)

Imagine you've trained a dog to do basic tricks (like "sit" and "stay"). Now you want it to perform a complex sequence of actions, like navigating an agility course. The dog knows the individual moves but doesn't know the correct order or what a "good" run looks like.

How do you train it?

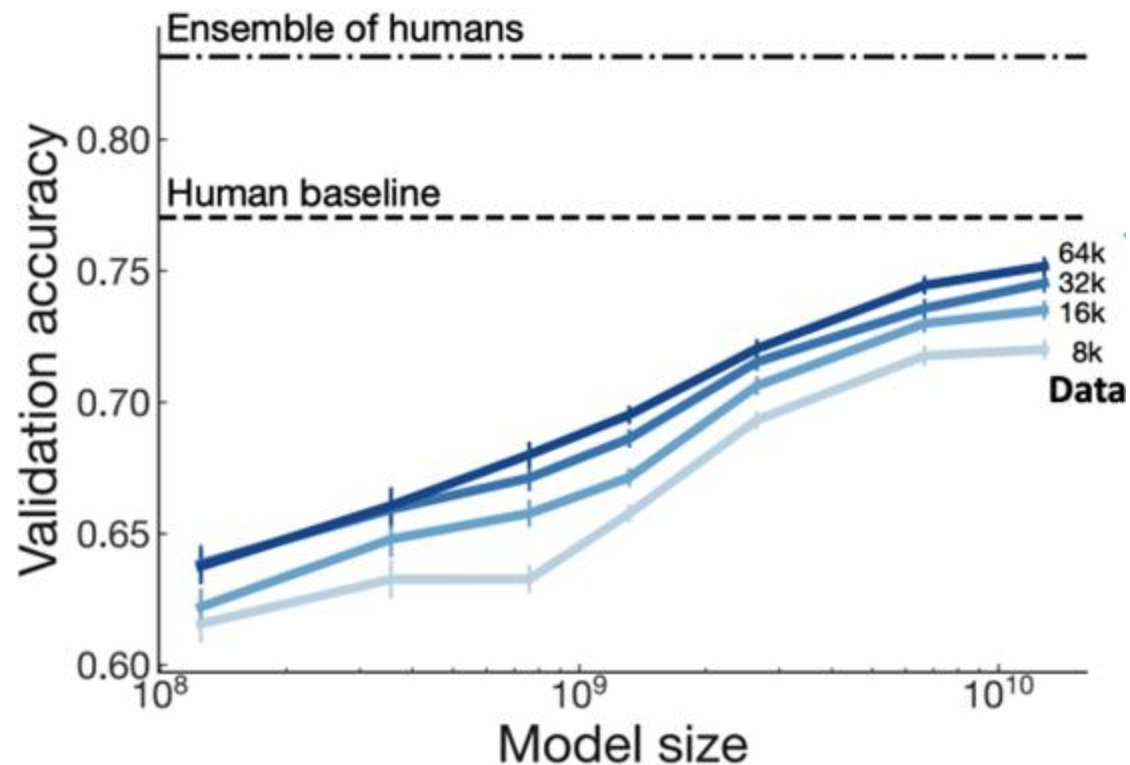
1. You **show it** what a perfect run looks like (demonstration).
2. The dog tries, and you **give it treats when it does something right** and correct it when it goes wrong (feedback).
3. Over time, the dog **figures out the exact behavior** that maximizes its treats.

RLHF is this same process, but for AI models. It's a technique used to finely align a language model's outputs with complex human values like **helpfulness, honesty, and harmlessness.**

RLHF

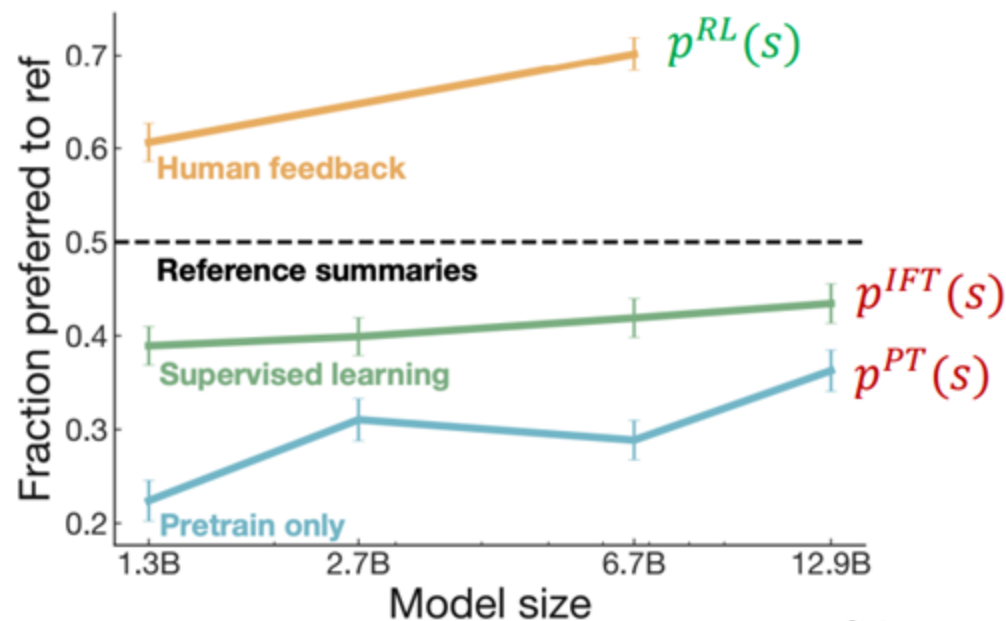
Make sure your reward model works first!

- Evaluate RM on predicting outcome of held-out human judgments

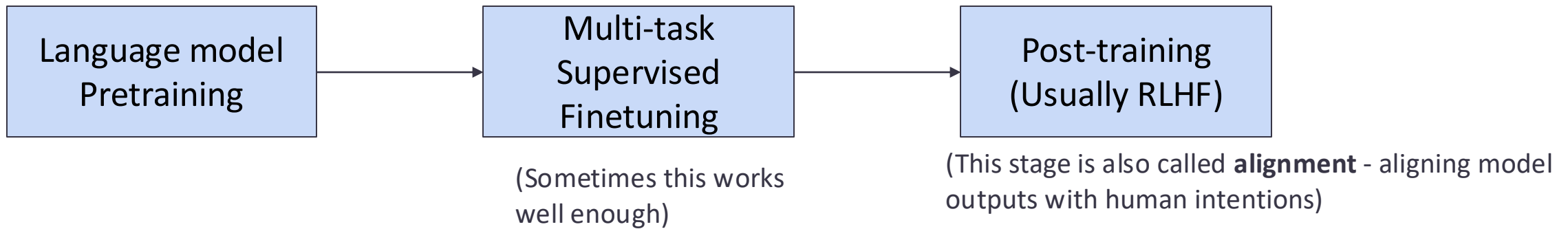


Large enough RM trained on enough data
approaching single human perf

RLHF provides gains over pretraining + finetuning



The Full LLM Pipeline



PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION

GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

After RLHF post-training, the LLM will follow human instructions much better

Limitations of RL + Reward Modeling

- Human preferences are unreliable!
 - **"Reward hacking"** is a common problem in RL
 - Chatbots are rewarded to produce responses that seem authoritative and polite, **regardless of truth**
 - They often reply "You are absolutely right!..." (instead of telling you that you are wrong), which is not helpful
 - This can result in making up facts + hallucinations

TECHNOLOGY

Google shares drop \$100 billion after its new AI chatbot makes a mistake

February 9, 2023 • 10:15 AM ET

<https://www.npr.org/2023/02/09/1155650909/google-chatbot--error-bard-shares>

Bing AI hallucinates the Super Bowl



Vision-Language-Models

Vision Foundation Models

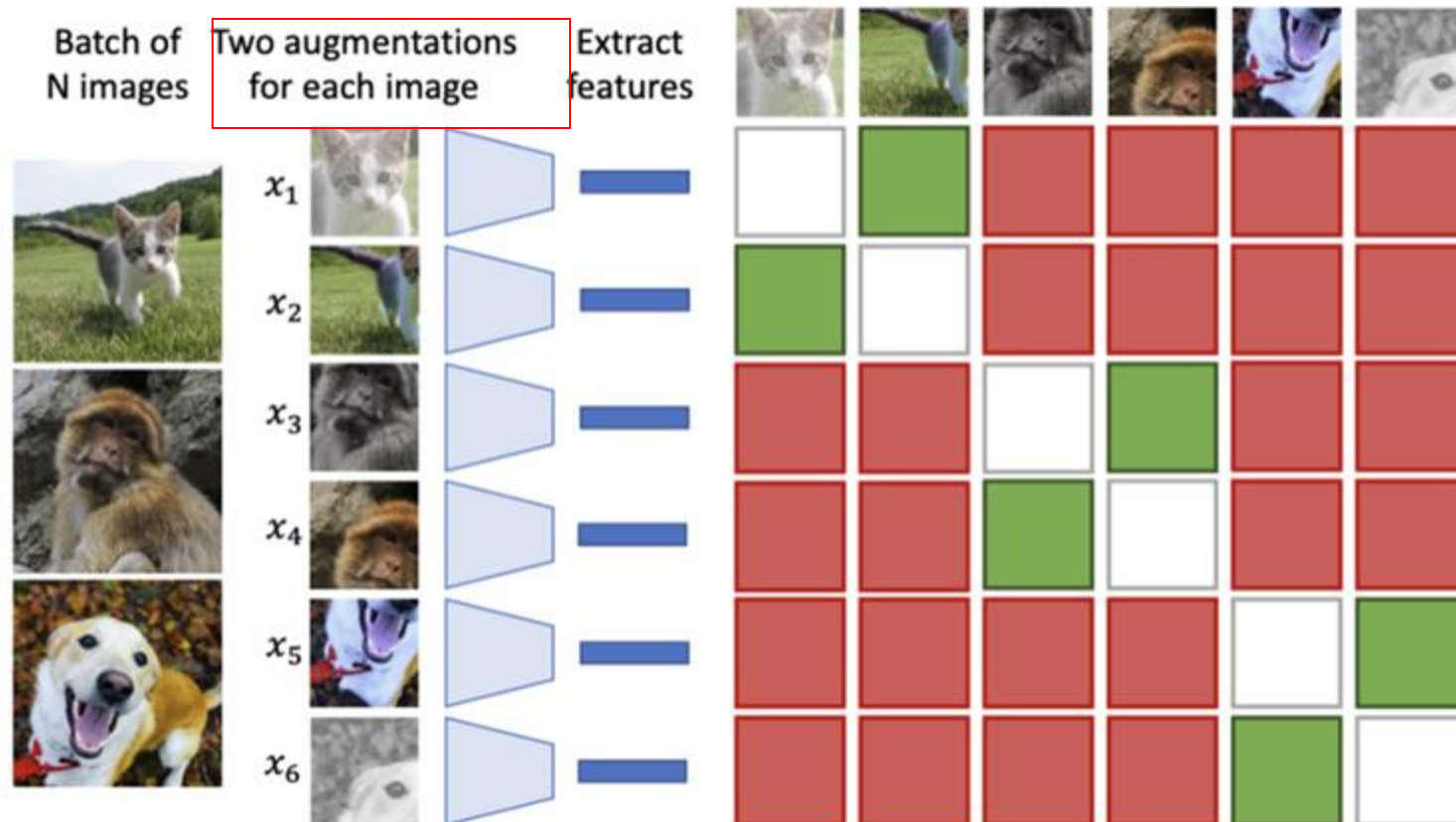
- There are many classes of Foundation models
 - We are likely to build on top of them for our downstream embodied AI tasks

<u>Language</u>	<u>Classification</u>	<u>LM + Vision</u>	<u>And More!</u>
ELMo	CLIP	LLaVA	Segment Anything
BERT	CoCa	Flamingo	Whisper
GPT		GPT-4V	Dalle
T5		Gemini	Stable Diffusion
		Molmo	Imagen

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CLIP: Contrastive Learning

Contrastive learning is often used to get good features. For example, SimCLR* uses **self-supervised learning** to learn good image features

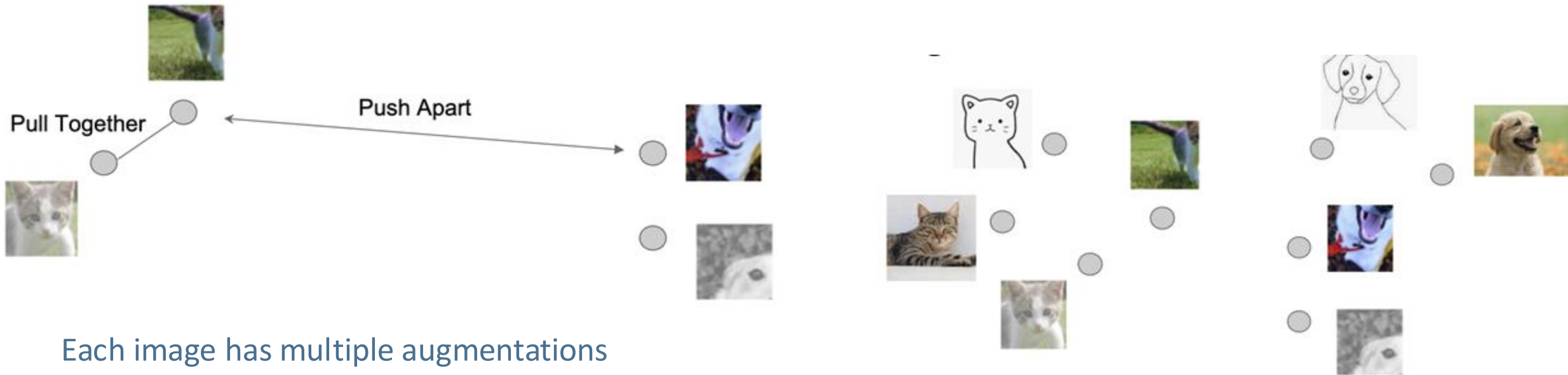


Can train small classifiers on top of these features using supervised learning

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CLIP: Contrastive Learning

The main idea was to learning concepts without labels -> a self-supervised pretraining objective

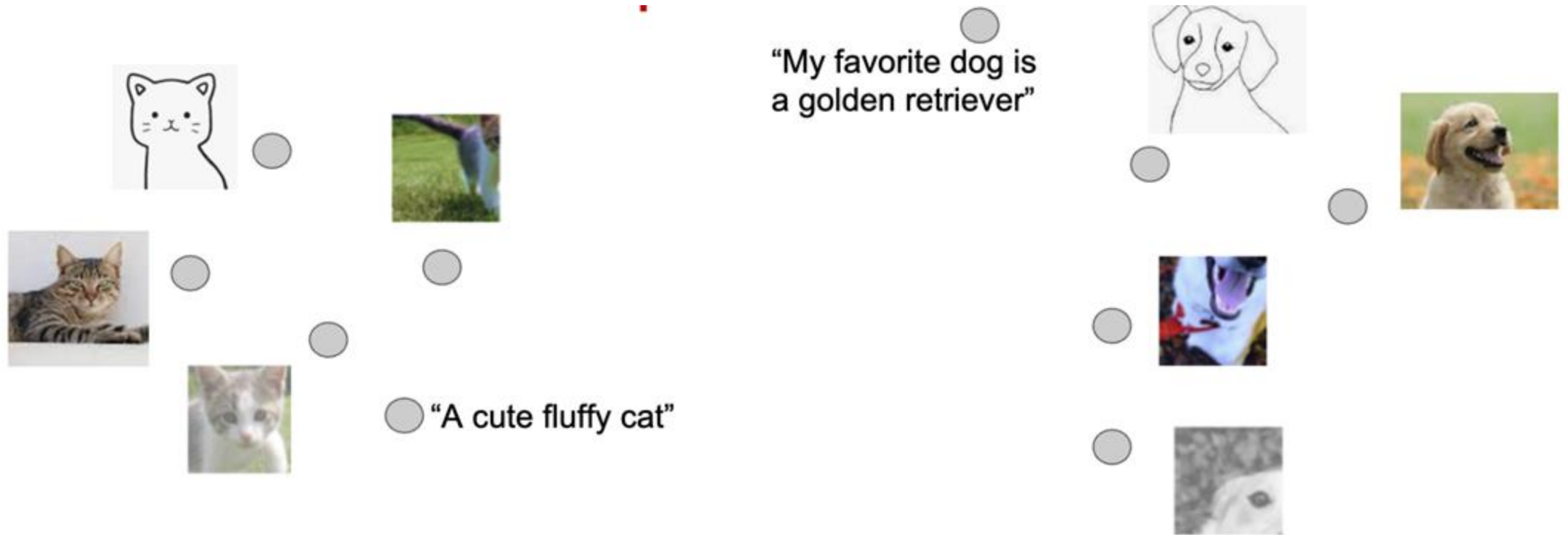


Each image has multiple augmentations (crops), so they are positives; Other images are negatives. You get large scale “free annotations”

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CLIP: Contrastive Learning

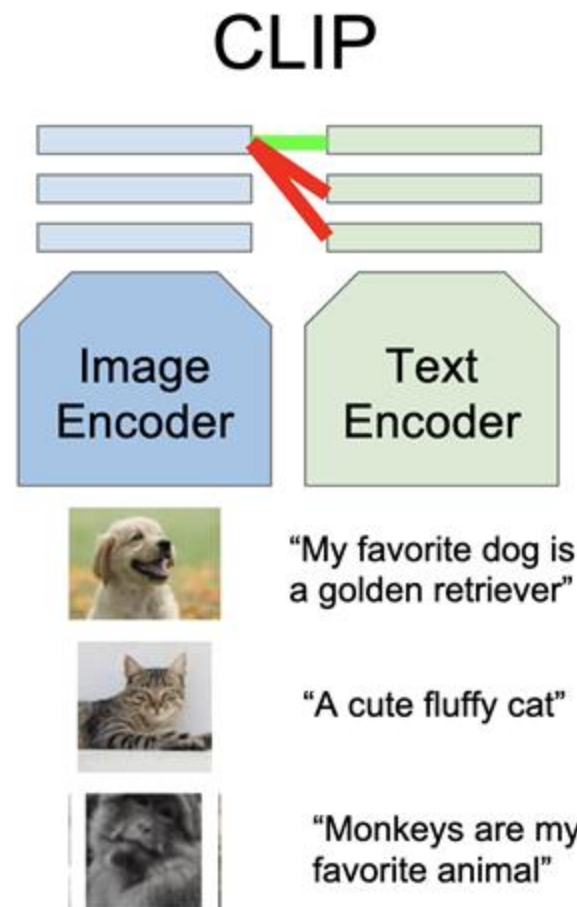
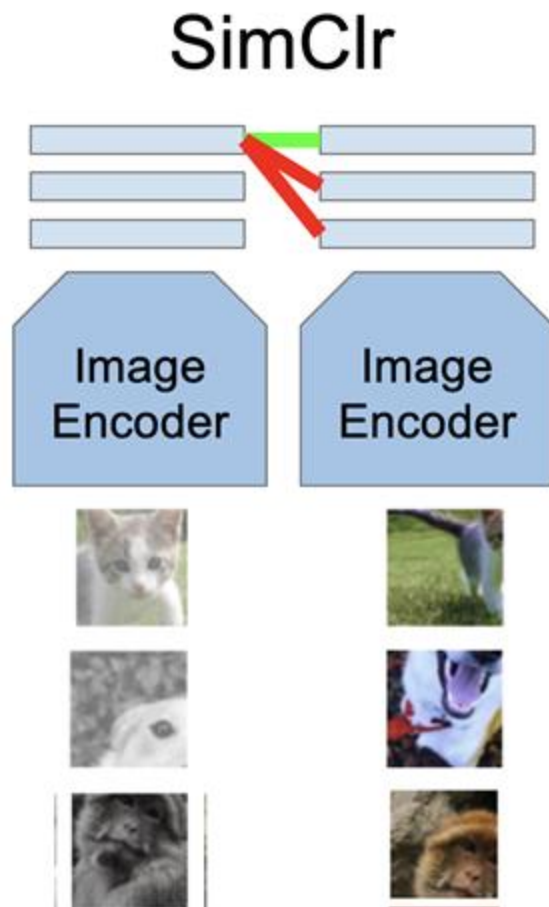
What if this representation space could also embed **sentences/phrases**?



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CLIP: Contrastive Learning

- Contrastive learning with image and text - CLIP



$$\sum_{i=1}^n -\log \left(\frac{e^{\langle u_i, v_i \rangle}}{\sum_{j=1}^n e^{\langle u_i, v_j \rangle}} \right)$$

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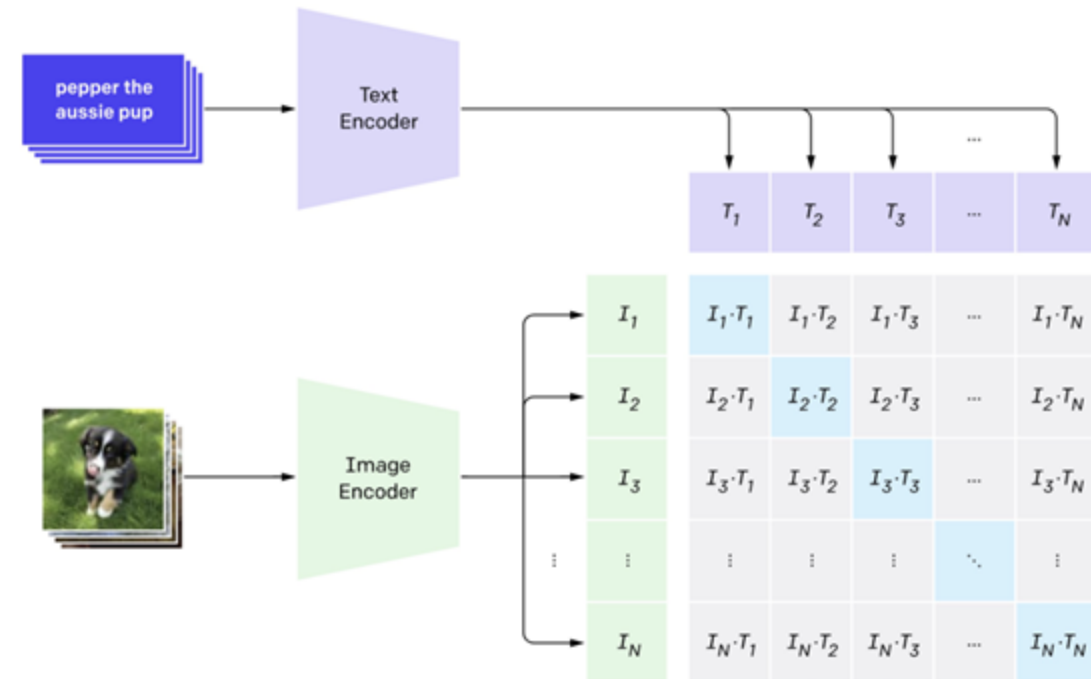
CLIP: Contrastive Learning

- Lots of image-text data can be found online



CLIP training data was scraped at scale from images and their associated alt-text from the internet. 400 million image-text pairs!

1. Contrastive pre-training

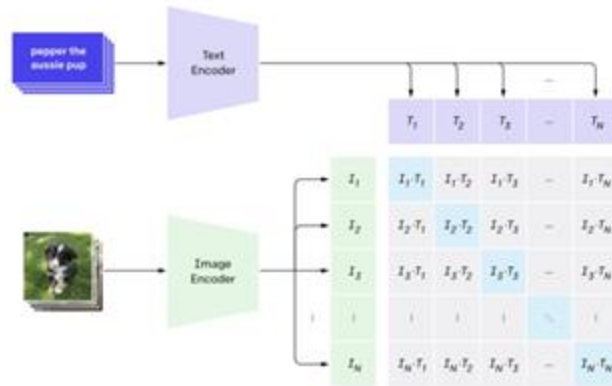


At the end of training, you have a model that will give you a similarity score between an image and text

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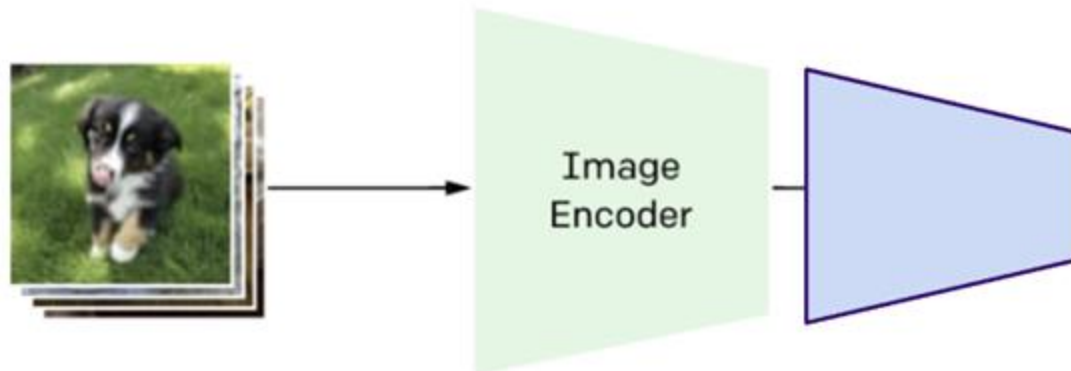
CLIP for Downstream Tasks

- Step 1: Pretrain a network on a pretext task that doesn't require supervision



Pre-training tasks:
Contrastive Objective

- Step 2: Transfer the encoder to downstream tasks via **linear classifiers**

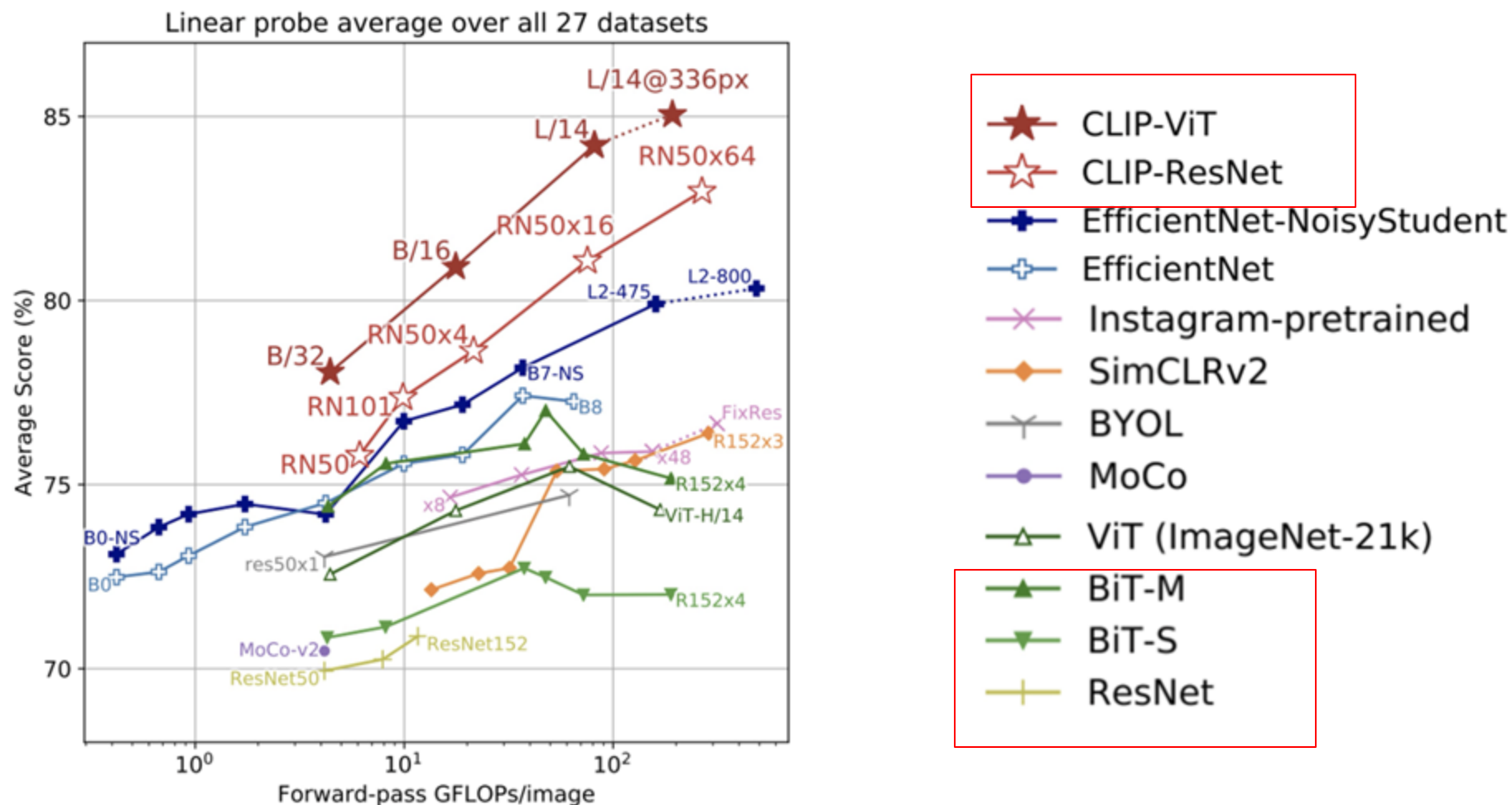


Downstream tasks:
Image classification, object
detection, semantic
segmentation

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CLIP: Contrastive Learning

- CLIP features **w/ linear probe** across multiple datasets

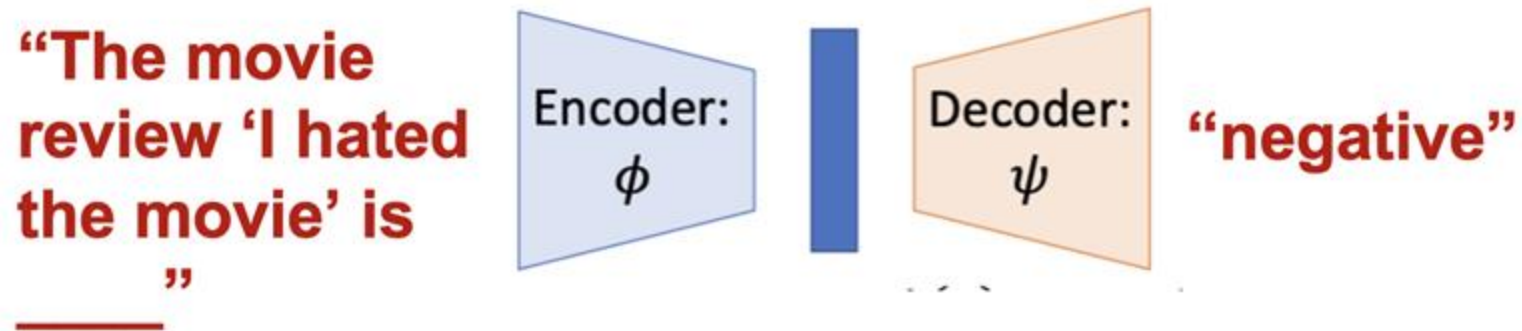
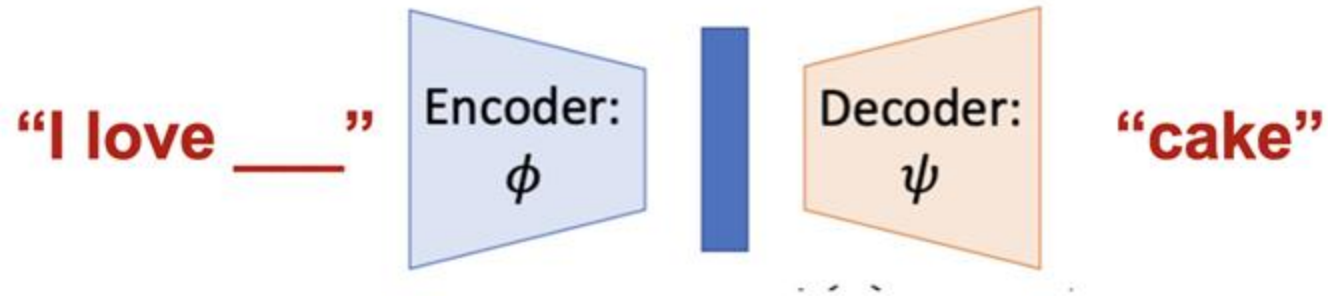


ViT vision backbone is better than ResNet backbone - recall the inductive bias we learn before.
10 points better than fully-supervised models like BiT.

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CLIP: How to Use It Zero-shot like LLMs?

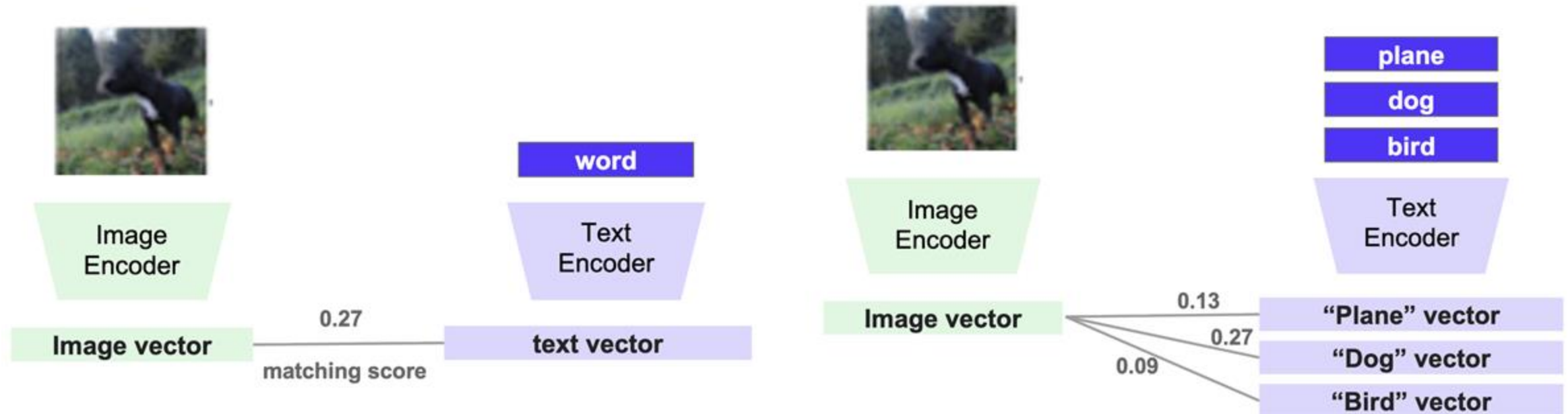
- For LLMs, we can use prompting to achieve zero-shot



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CLIP: How to Use It Zero-shot like LLMs?

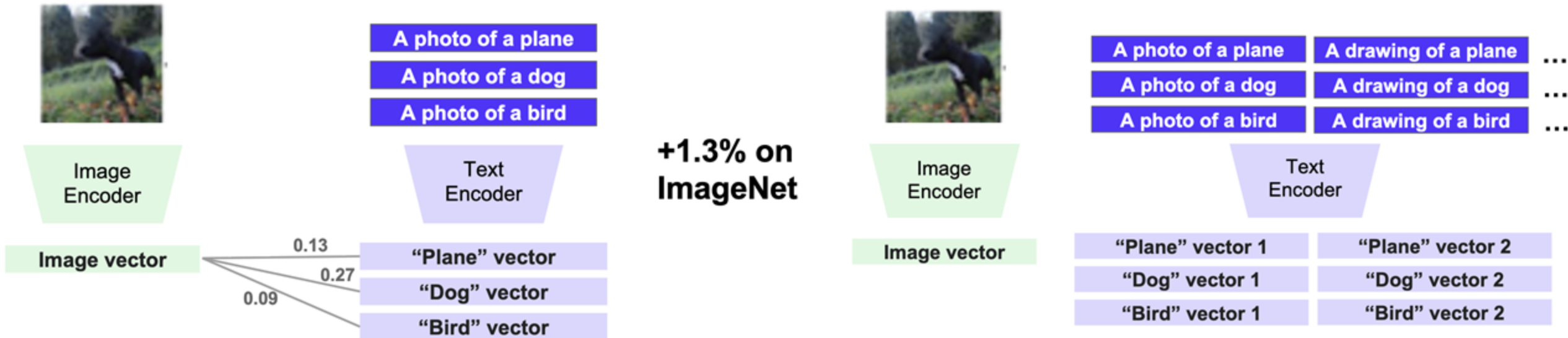
- CLIP: we can create a classifier using the text encoder
- Match a new image to the most similar vector



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CLIP: How to Use It Zero-shot like LLMs?

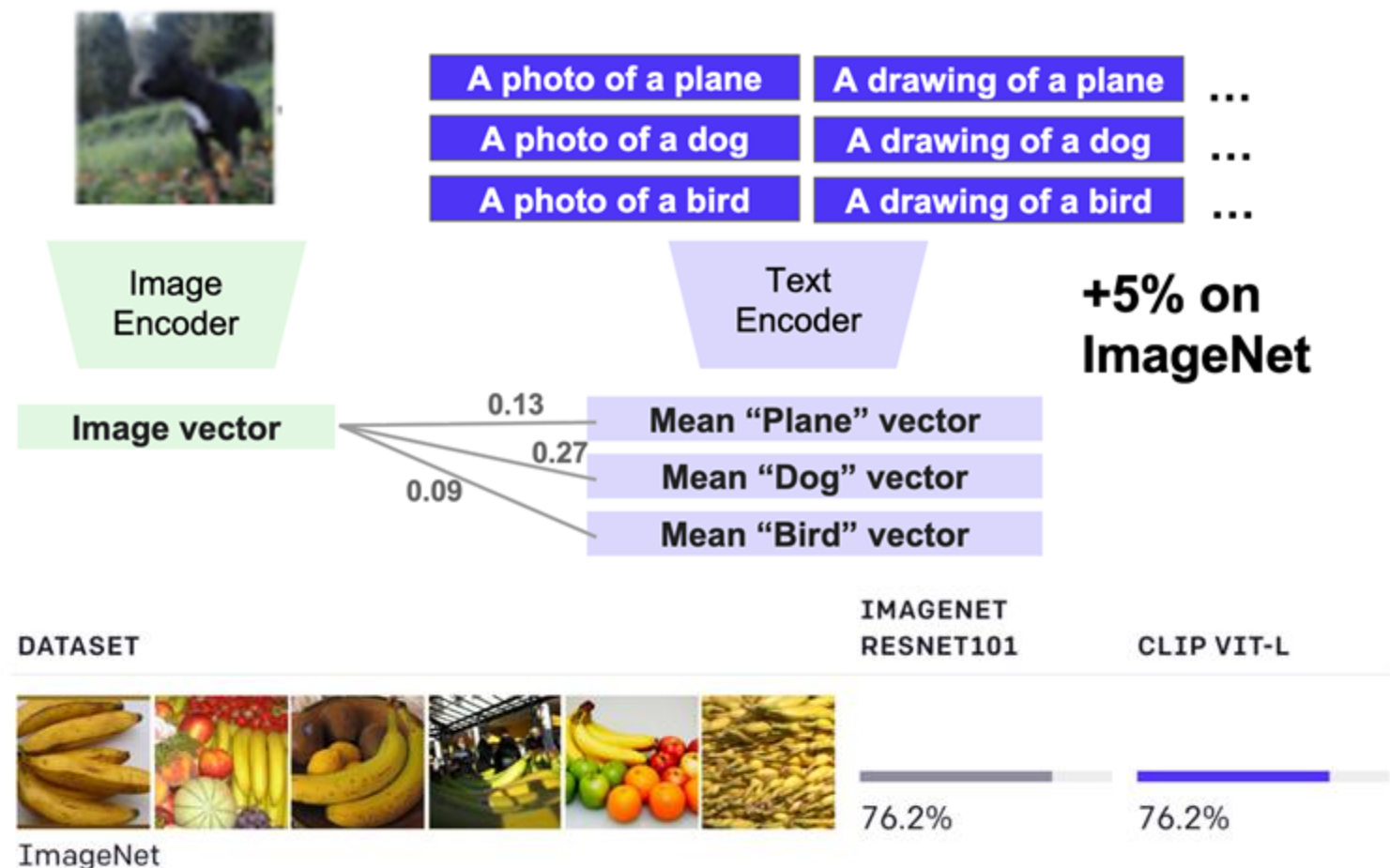
- CLIP: we can create a classifier using the text encoder
 - Since CLIP was trained with phrases, you can improve performance by **using a phrase “A photo of a [category]”**
 - A single phrase might be too biased. Solution: **Use multiple phrases**



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CLIP: How to Use It Zero-shot like LLMs?

- CLIP: we can create a classifier using the text encoder
 - Use the average vector across phrases as their presentation for each category

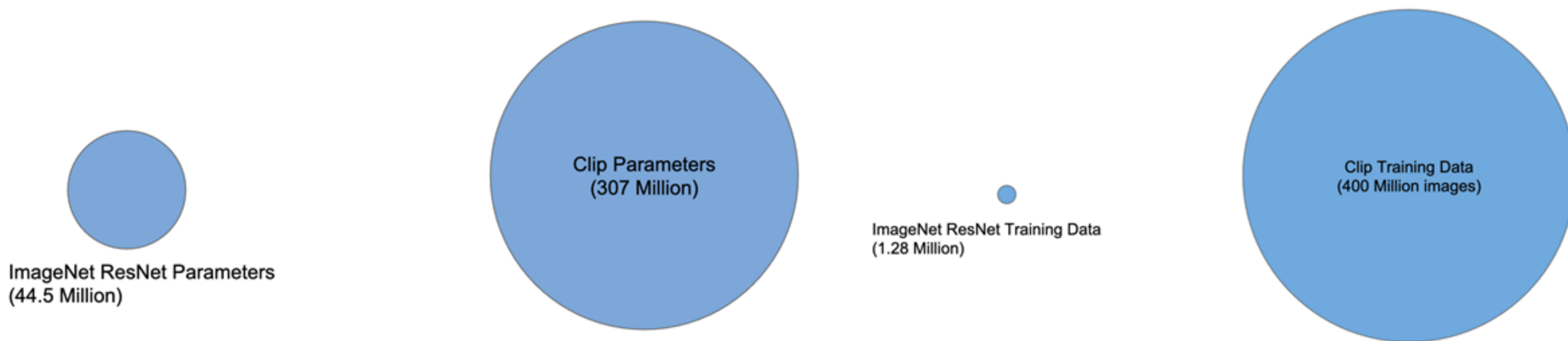


Exciting result after training on 400M image-text pairs
Matches the accuracy of of ResNet 101 that has been trained on ImageNet, except CLIP was trained with no human labels at all!

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CLIP: Zero-shot Performance Matches Fully-Supervised

- CLIP scaled up the **model parameters** with the transformer architecture
- CLIP scaled up the training data by scraping image-text pairs from the internet



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Problems with the CLIP model

- Only understands “compositional semantics” - It relies too heavily on batch size to learn concepts
- Even in a batch of 32K, it’s unlikely you see both “a mug in some grass” and “some grass in a mug”

CLIP can’t distinguish between:



there is a mug in some grass



there is some grass in a mug



an elderly asian woman wearing a straw-like hat sits outside near a bicycle while a gray car is about to pass by.

These get similar scores

an like gray hat wearing a bicycle - asian woman sits outside near a straw while a about car is elderly to pass by.

Vision-Language-Models

Problems with the CLIP model

- Only understands “compositional semantics” - It relies too heavily on batch size to learn concepts
- Solution? Build hard negatives. But training with hard negatives has its own problems...



A black cat and a brown dog
✓

A brown cat and a black dog
✗

A brown dog and a black cat
✗

Vision-Language-Models

Problems with the CLIP model

- Image-level captions are insufficient supervision



“living room”

✓

“house plants”

X

“couch”

X

Vision-Language-Models

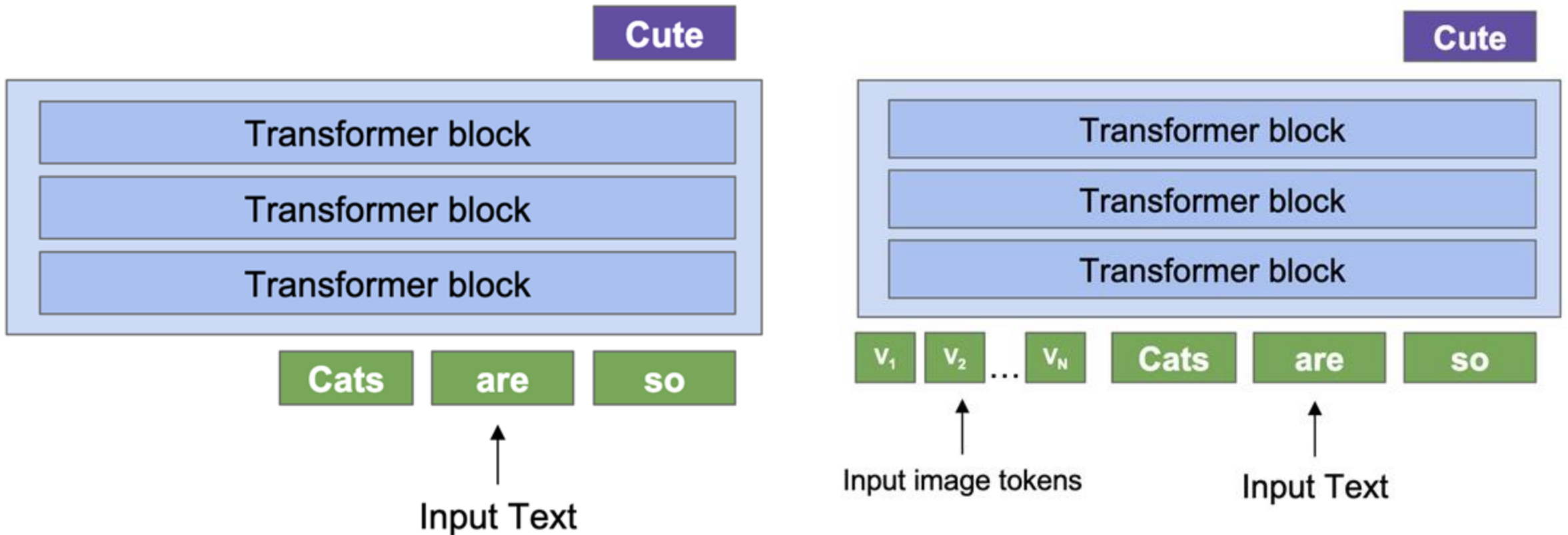
LLaVA - Generative Vision-Language Model

- Motivation:
 - CLIP-like contrastive learning leads to bad semantic capabilities
 - Language models which do next token prediction can be applied to a wide variety of tasks at inference (Math, sentiment analysis, symbolic reasoning)
 - Can we build a model that can accept images and text as input, and then output text?
 - => Vision-Language Model!

Vision-Language-Models

LLaVA - Generative Vision-Language Model

- Recall how transformers decode language
- Key idea behind LLaVA – add visual information to the LLM

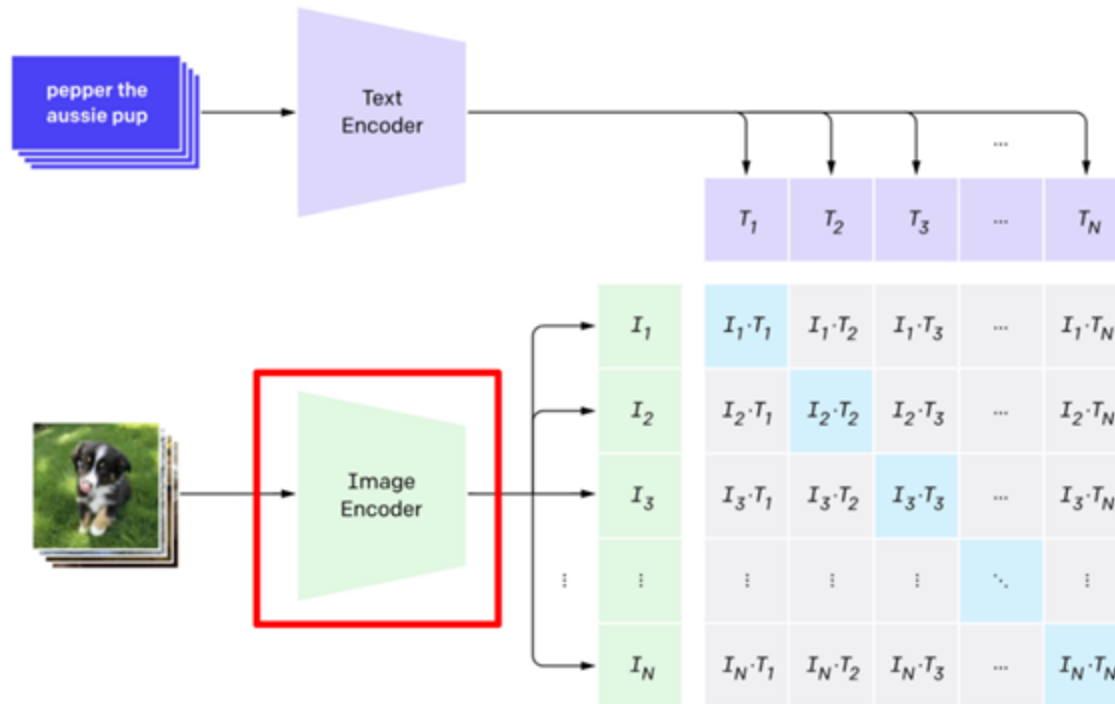


Vision-Language-Models

LLaVA - Generative Vision-Language Model

- Image tokens: The CLIP encoder is a good option!

1. Contrastive pre-training

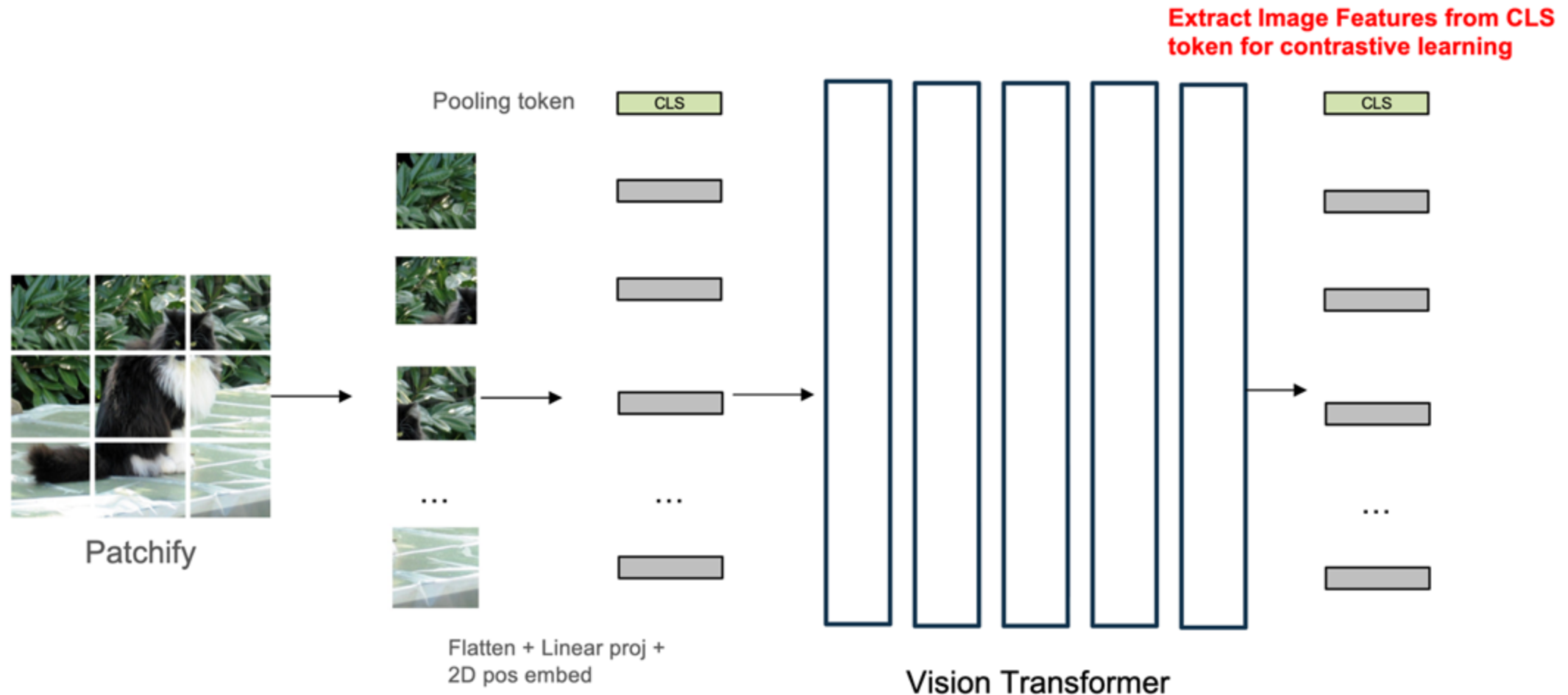


At the end of training, you have a model that will give you a similarity score between an image and a text

Vision-Language-Models

LLaVA - Generative Vision-Language Model

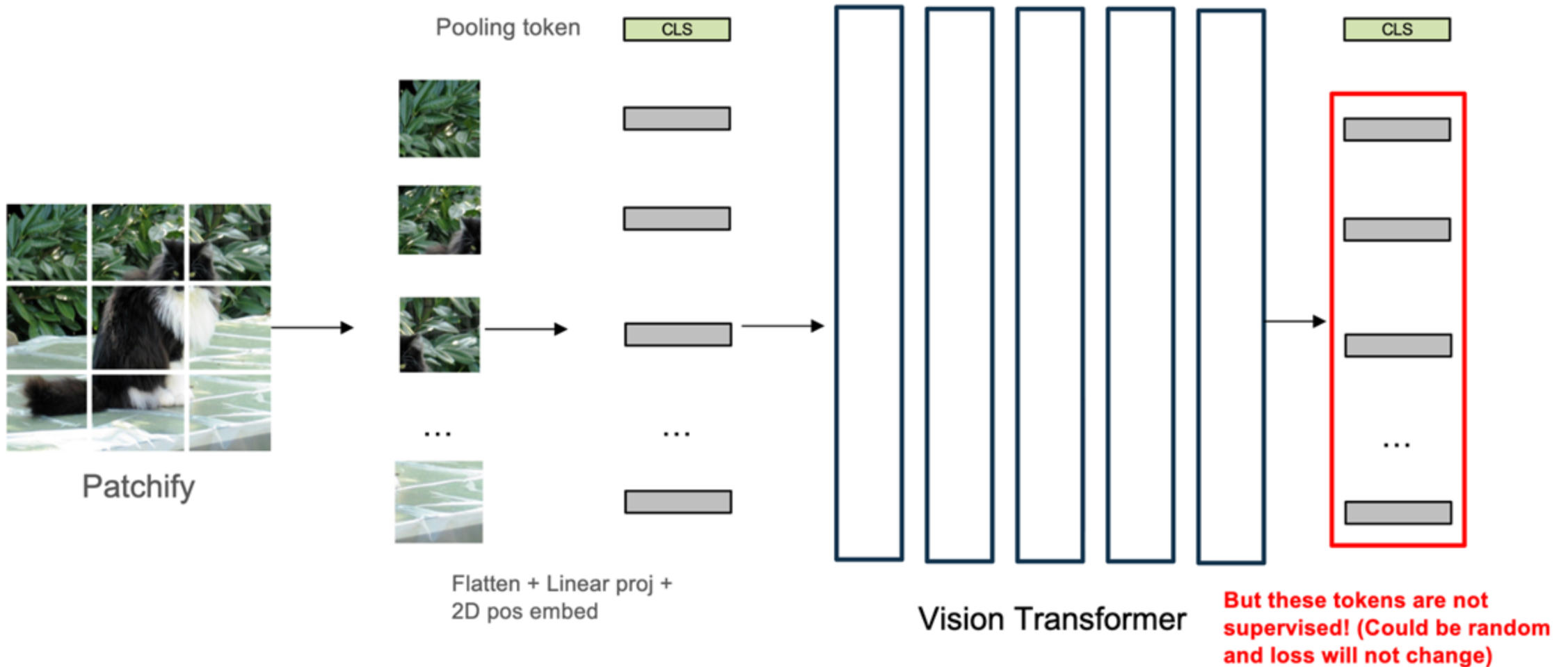
- What features should we use from CLIP?



Vision-Language-Models

LLaVA - Generative Vision-Language Model

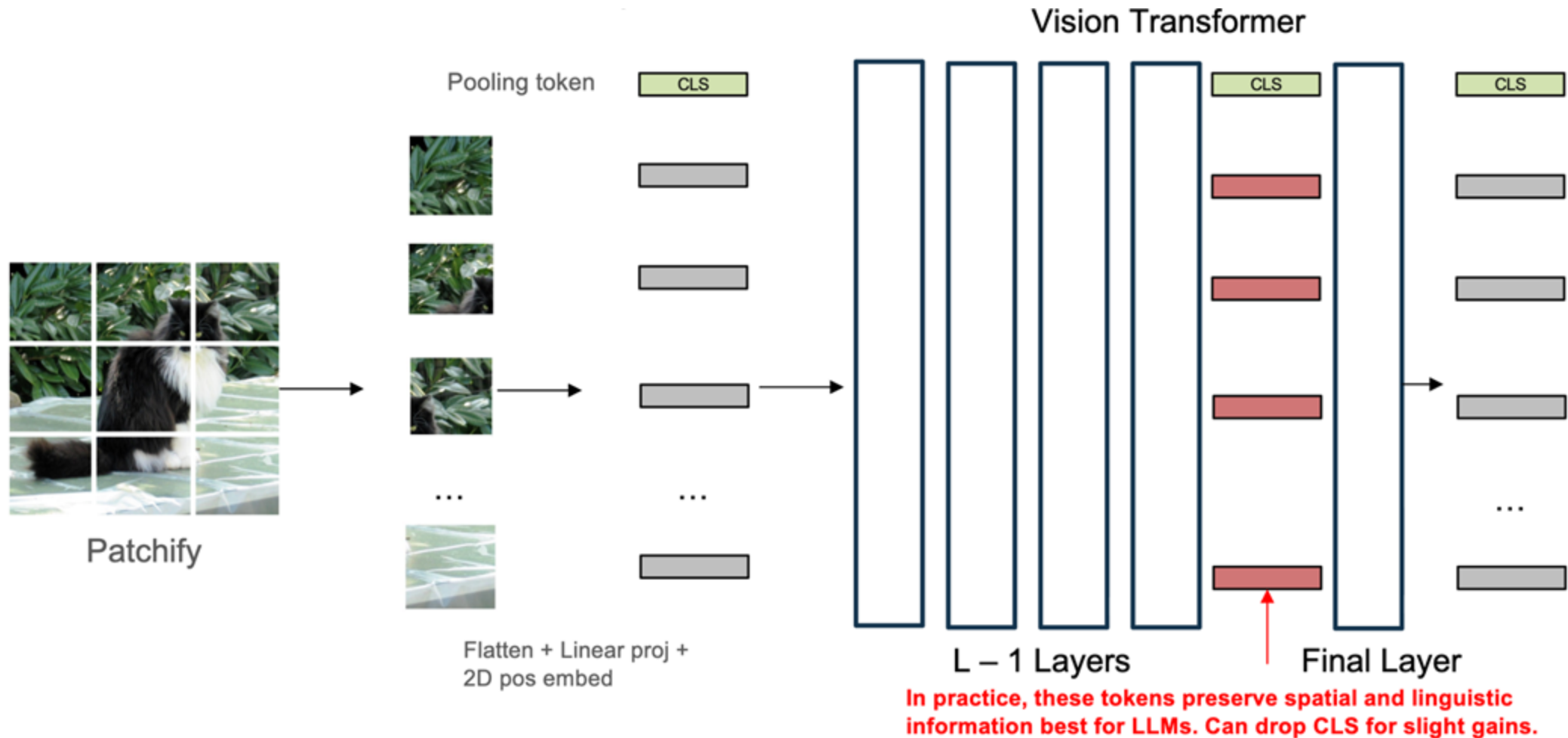
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LLaVA - Generative Vision-Language Model

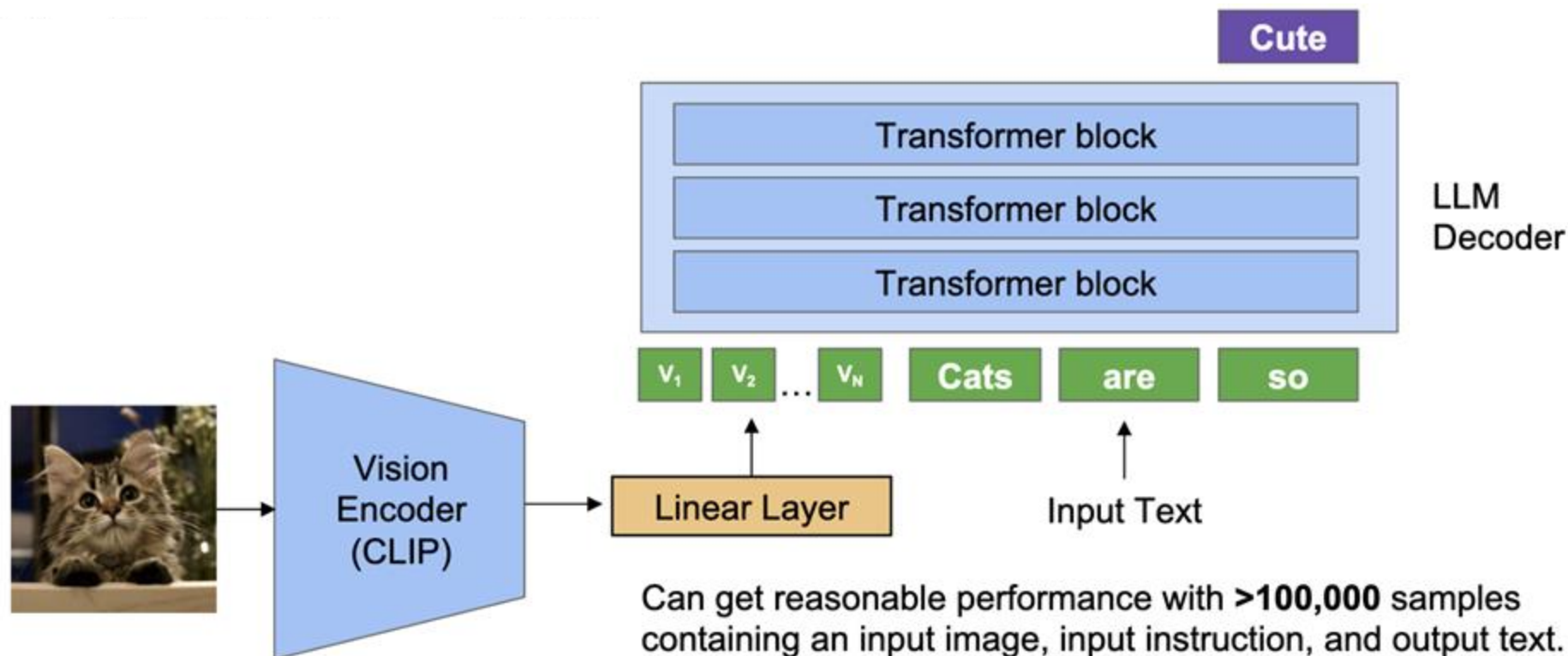
- Use the last - 1 layer!



Vision-Language-Models

LLaVA - Overall Architecture + Training Recipe

- 1. Initialize with pretrained Language Model for LLM Decoder (e.g. LLaMA) and pretrained image encoder (e.g. CLIP)
- 2. Train a new linear layer to bridge CLIP features to LLM input space
- 3. Finetune LLM + linear layer together

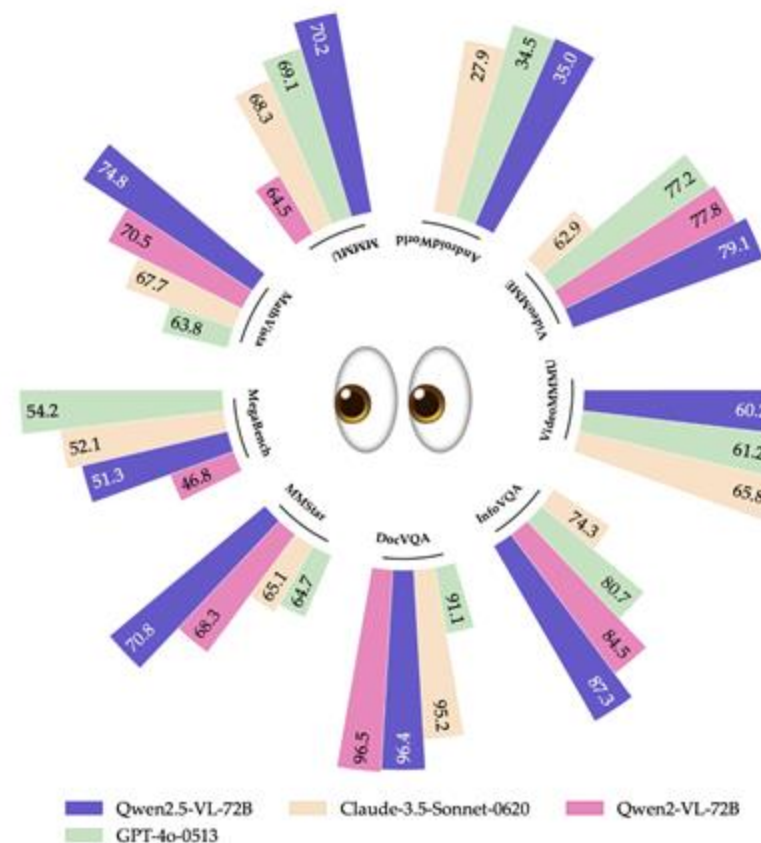


Vision-Language-Models

Qwen-VL - Currently the best open-source VLM

- Qwen2.5-VL and Qwen3-VL are the go-to VLMs as of Nov. 2025
- You can use tools like vLLM to start a fast VLM server for your application
 - You can achieve ~30 token/s with 2x4090 GPUs (70k RMB) for Qwen2.5-VL-32B model
 - And 7B model on 3090 machine (10k RMB)

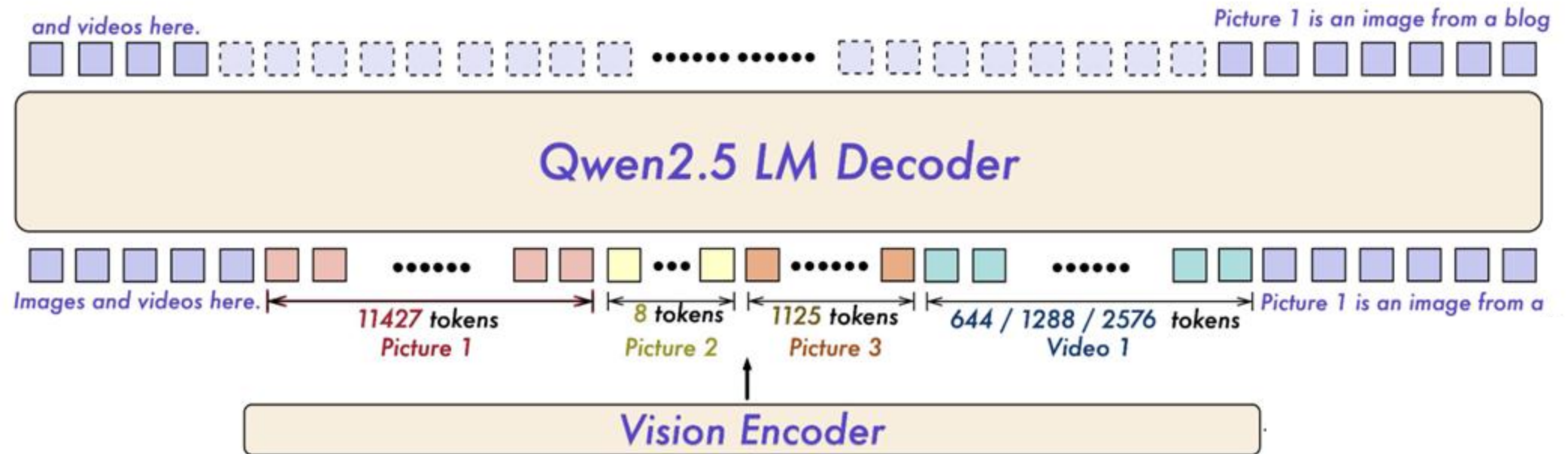
Much better than GPT-4o
on DocVQA etc.



Vision-Language-Models

Qwen2.5-VL - Architecture

- Similar to LLaVA, putting vision tokens interleave with text tokens for the decoder



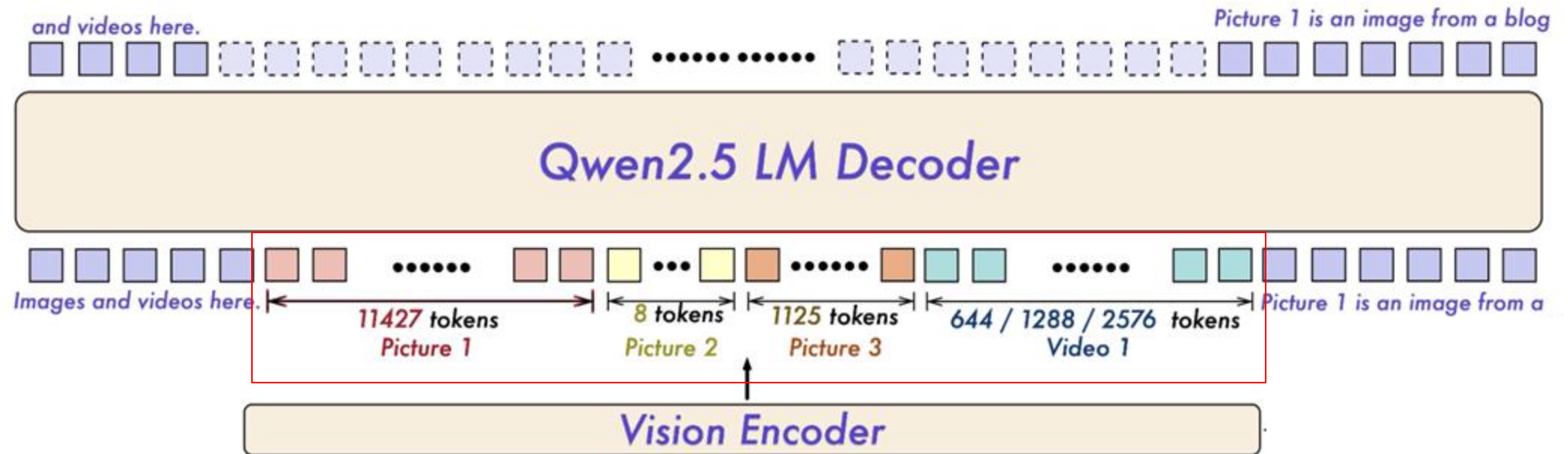
The vision encoder is still a (modified) ViT model. But they claim it is **trained from scratch**.

“The training process consists of several stages, including **CLIP pre-training**, vision-language alignment, and end-to-end fine-tuning.”

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Qwen2.5-VL - Architecture

- Similar to LLaVA, putting vision tokens interleave with text tokens for the decoder

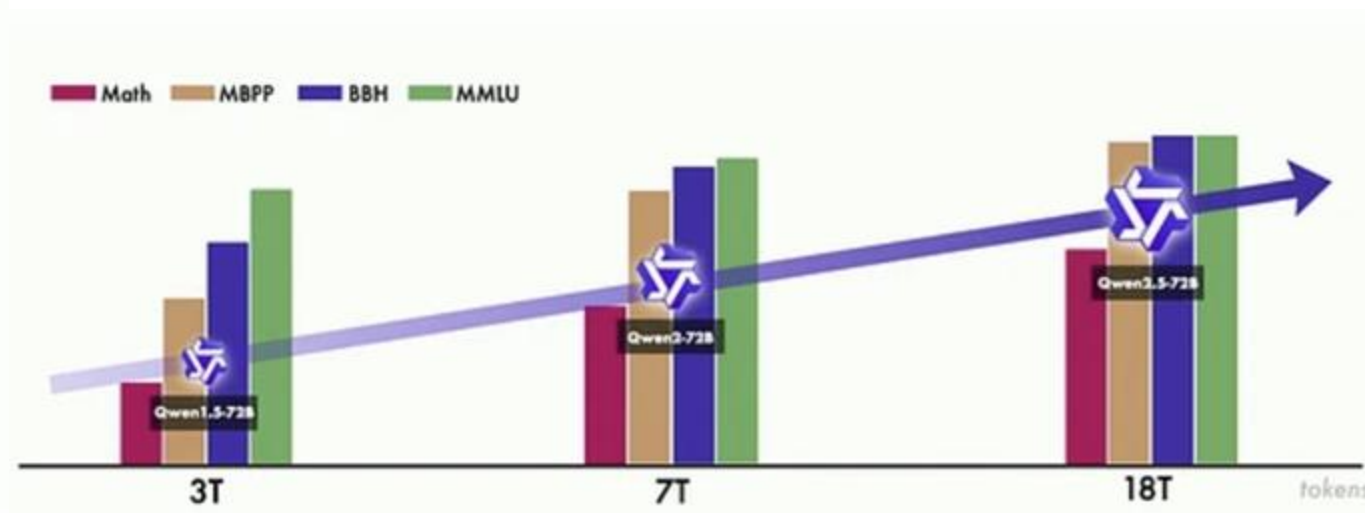
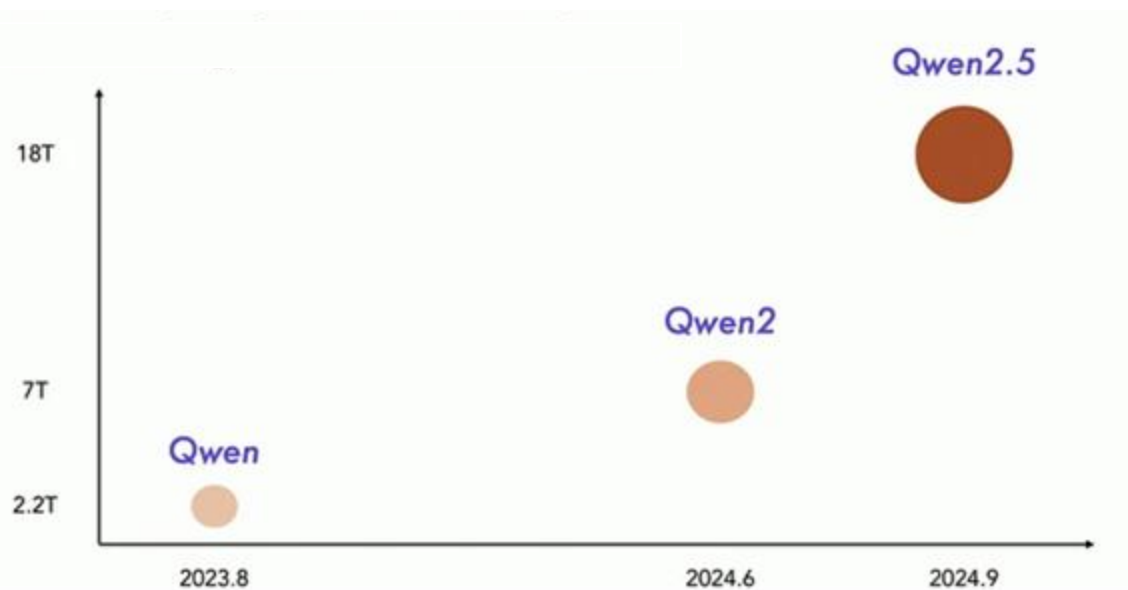


Notice here, different resolution images will take up different number of tokens.
It also supports video inputs (consider them as image token inputs with timestamps)

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Qwen-VL - Why it is better than LLaVA

- Scaling data quantity and quality efficiently
 - Data scaling: from 7T to 18T tokens
 - Model performance scales with data (for the same 72B model)



Vision-Language-Models

Qwen-VL - Why it is better than LLaVA

- Scaling data quantity and **quality** efficiently
 - Good filters are what you always need
 - Better math and code data

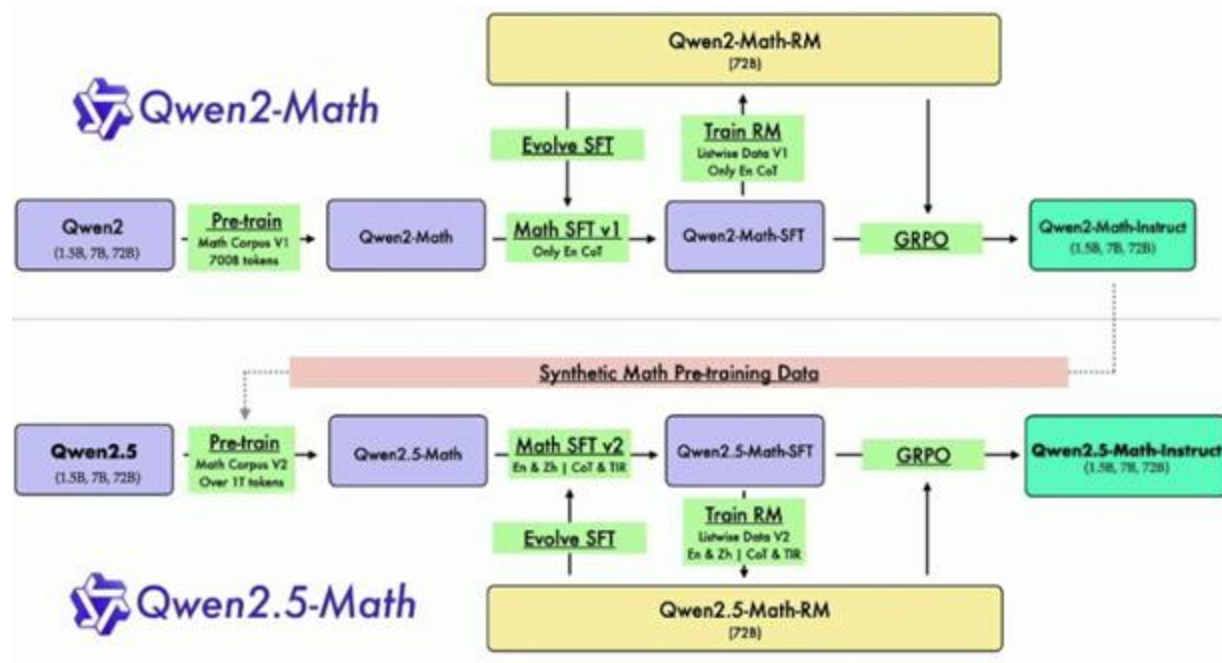
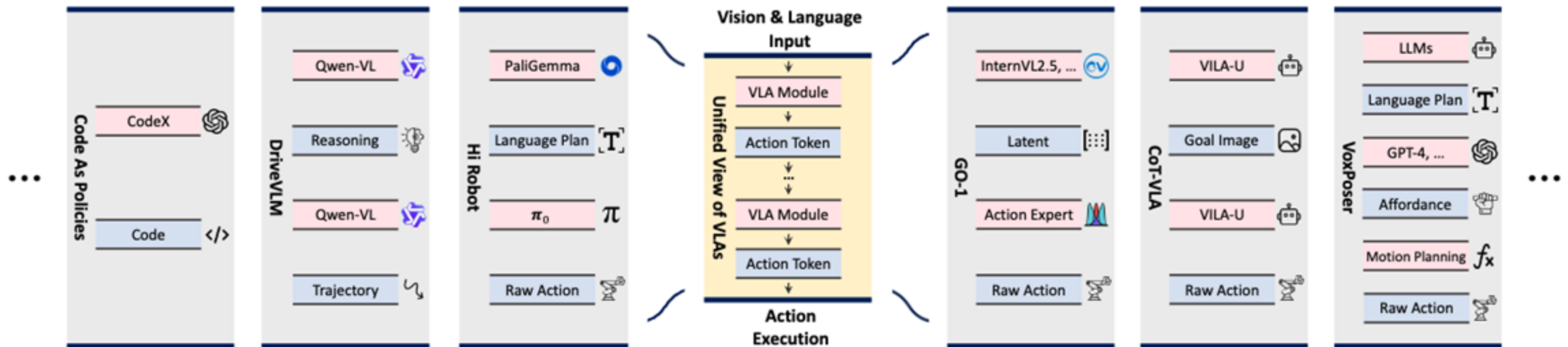


Figure 2: The three-stage training pipeline for Qwen2.5-Coder.

Vision-Language-Models

From LLM to VLM to VLA

- For embodied AI tasks, the outputs we want are **action tokens**
- The inputs are similar (vision + text tokens) with LLM and VLM
 - Plus robot states inputs
- Tremendous progress in language and multimodal (vision+language) models
- We can leverage these to improve capabilities to learn new things



Project 3

This is a reminder that the deadline for **Project 3 is November 24**. **To stay consistent with the other class, the deadline will remain unchanged.**

The period after November 24 will be reserved for preparing your presentation slides (PPT).

If you have any questions about your code, please bring your laptop to class; we will be available after class to help you debug and answer questions. Thanks!



Thanks for your attention!

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